

Power Budget / Bidirectional Coupler

Laboratory Experiment #6

By

Katherine Mullins

Submitted To:

Professor Radu Bunea

**ETS 2220C- Introduction to Fiber Optics
Department of Electrical and Computer Engineering Technology A.S.
Division of Engineering, Computer Programming, and Technology
Valencia College - West Campus**

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Lab Report

1.) Objective

The objective and intent of our group in this experiment was to do power budget calculations through a multimode bidirectional coupler.

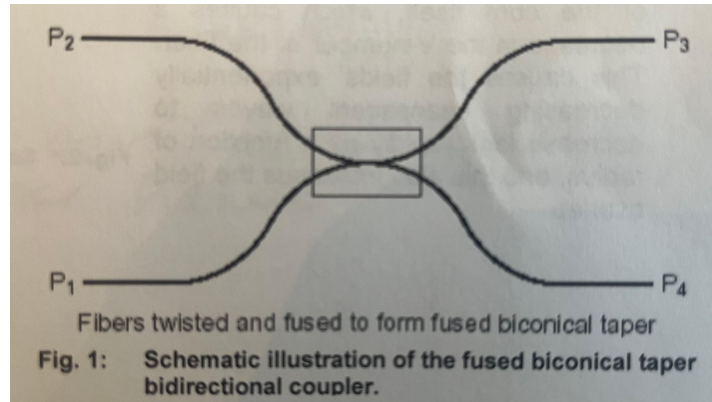
2.) Materials

- Optical Table (SG-22-2)
- HeNe Laser
- Rod Post (45)
- Adjustable Mounting Fixtures for HeNe Laser
 - Laser Mount (ULM-TILT)
 - Rod Clamp (340-RC)
- Fiber Coupler (F-916)
- 1.5m Fiber Optic Multimode (F-MLD)
- Bidirectional Coupler
- Fiber Positioner (FP-1)
- Bare Fiber Holder (FP3-FH1)
- Bare Fiber Holder Mount (818-FA2)
- Sumitomo Cleaver (FC-6M)
- Sumitomo Stripper (JR-22)
- Objective Lens (M-40x)
- Post Holder (VPH-2)
- Post (SP-2)
- Goggles
- Aluminum Square Rule
- Tape
- Ruler & Meter Stick
- Optical Detector (1815-C)
- Optical Power Meter (818-SL)
- Screw Kit, $\frac{1}{4}$ x 20 (SK-25A)

3.) Background

In this project, the components which are needed to build a fiber data distribution system will be explored. Fiber optic components for data distribution are of importance in short-haul applications, such as local area networks (LANs), which provide information transfer within structures such as office buildings and manufacturing facilities, and in such vehicles as aircrafts and ships. These components are also used in longer-haul systems in order to fully utilize the bandwidth of the optical fiber.

The need for devices to combine light from several fibers into a single fiber or to split the light from one fiber into two or more fibers has been recognized ever since optical fibers were first considered as a communication medium. A 2X2 bidirectional coupler, which is a device for splitting the light in one input fiber into two output fibers is illustrated schematically in Figure 1.



In fiber optic systems, they play a role which is equivalent to that of beam splitters in conventional optics or T-taps in electronics. Bidirectional couplers play a critical role in fiber optic instrumentation, high-data-rate LANs, and communication systems. Star couplers are nxn devices of the same basic construction.

Bidirectional couplers can be classified into two broad categories, those that use conventional optical beam splitting techniques and those that couple energy between the two fibers using the overlap of the evanescent waves of the light in the cladding of the fibers. Several designs using beam-splitting techniques have been developed. These have not found widespread use because of the problem of maintaining the alignment of discrete micro-optic components under extreme conditions. However, this type of device has an advantage over the evanescent wave coupler in that the modal content of the input fiber is preserved in the output fibers.

When the cores of two fibers are close together, the energy is shared between the two overlapping evanescent waves and energy is transferred from one fiber to the other. For this reason, energy coupling between two fibers by this technique is known as evanescent field coupling. The evanescent fields will penetrate significantly into the cladding only for the highest-order rays, i.e., those nearest the critical angle for total internal reflection. Because of this, the coupling obtained using this technique is highly dependent on the modal content of the input fiber. If the light in the input fiber is carried in predominantly higher-order modes, more of the light will be coupled to the second fiber. If the light is predominantly lower-order modes, less light will be coupled. Bidirectional couplers which use this technique actually achieve their specified coupling ratio when light is evenly distributed among the allowed modes of the input fiber.

For instance, light which comes out of port 3 of a bidirectional coupler will be in mostly higher-modes while the light remaining in the input fiber coming out of port 4 will be in lower-modes. This will also have an effect on the performance of any other bidirectional

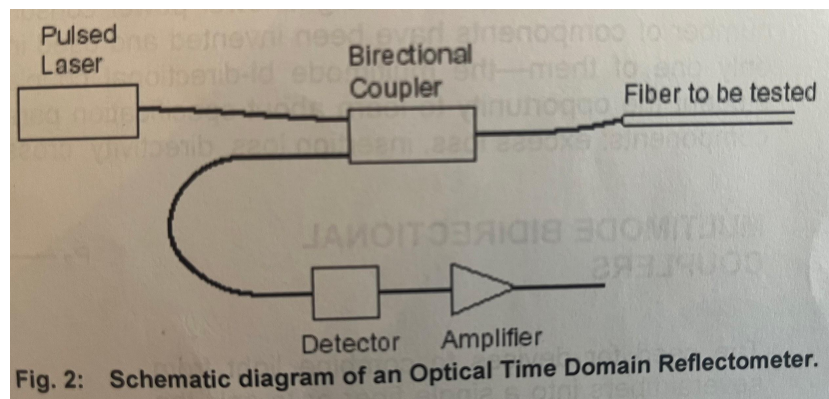
couplers which are now connected in series with the first coupler since the distance between couplers is not likely to be great enough to allow the light to be redistributed among the modes. Couplers downstream from port 3 will split off more than the specified power, while couplers downstream from port 4 will split off less than the specified power.

Because of the exponential decay of the evanescent field, the two fiber cores must be brought in close proximity for electric field overlap and optical power coupling to occur effectively. The most common technique for fabricating the coupler is called the fused biconical taper technique. In this

technique, the two fibers are twisted together and then heated to about 1500°C. While the fibers are being heated, they are subjected to a tension which causes them to stretch. This stretching increases the coupling between the two cores by two mechanisms. The first is due to the fact that when the fiber is stretched, the fiber cladding is tapered and becomes thinner, causing the two cores to come closer together, thereby increasing the evanescent field overlap. The second comes about because of the thinning of the core itself, which causes a decrease in the V-number of the fiber. This causes the evanescent field waves to decrease less rapidly as a function of radius, increasing the field overlap.

In practice, the stretching is continued until the desired coupling ratio is achieved. When the heating is stopped, the core separation is permanently locked in place. Couplers made by the fused biconical taper technique are extremely insensitive to temperature changes.

The bidirectional coupler which results was shown in Figure 1. Light which is input at port 1 may either come out of the same fiber at port 4 or be coupled to the second fiber and emerge at port 3. A small amount of light may be returned to the input port of the second fiber, port 2, because of backscattering and reflections due to the constriction at the taper.



The Coupling Ratio or Splitting Ratio is defined as the power coupled across to the second fiber divided by the total power through the coupler:

$$CR = -10 \log \left[\frac{P_3}{P_3 + P_4} \right]$$

Typical coupling ratios for commercially available devices are 3 dB (50%), 6 dB (25%) and 10 dB (10%).

The **Excess Loss** is defined as the total output power divided by the total input power,

$$EL = -10 \log \left[\frac{P_3 + P_4}{P_1} \right]$$

Early couplers exhibited losses in the range of 2–3 dB. Most couplers today exhibit losses in the range of 0.5 – 1.0 dB or less.

The **Directivity Loss** of the coupler is defined as the power returned to the second fiber on the input side divided by the input power,

$$DL = -10 \log \left[\frac{P_2}{P_1} \right]$$

This ratio is typically about –40 dB for good couplers.

The **Insertion Loss** of the coupler is defined in terms of a power ratio at an output port relative to the input, i.e.

$$IL_3 = -10 \log \left[\frac{P_3}{P_1} \right] \quad IL_4 = -10 \log \left[\frac{P_4}{P_1} \right]$$

4.) Procedure

Laser Alignment:

1. Mount the post at the end of the optical table. Attach the alignment fixture onto the post using ¼ - 20 socket screws. Mount the HeNe laser head/assembly so that the anticipated output of the beam will be along a hole pattern of the optical table.
2. Carefully mount an unplugged HeNe laser head into mounting/alignment fixture.
3. Connect the laser with the power supply. Power ON and ensure that the laser aperture is closed.
4. Mount the Fiber Coupler on the same axis in front of the laser (5").
5. Unblock the laser.

6. Adjust the alignment fixture position until the beam strikes through the center of the fiber coupler.
7. Using a ruler, determine if the laser is in alignment with the fiber coupler.
 - a. Measure the height between the optical table and the center of the fiber coupler.
 - b. Measure the height between the optical table and the laser beam in front of the laser.
 - c. Measure the height between the optical table at the end of the table.
 - d. Adjust the HeNe laser using the knobs from the mount until all measured heights are the same.
8. Mount a piece of paper at the end of the optical table.
9. Place the objective lens in the fiber coupler and using the knobs marked in the picture below, adjust until a perfect laser beam circle is projected on the paper.
10. Block the laser.
11. From here on, do not adjust the laser.

Laser / Fiber Alignment:

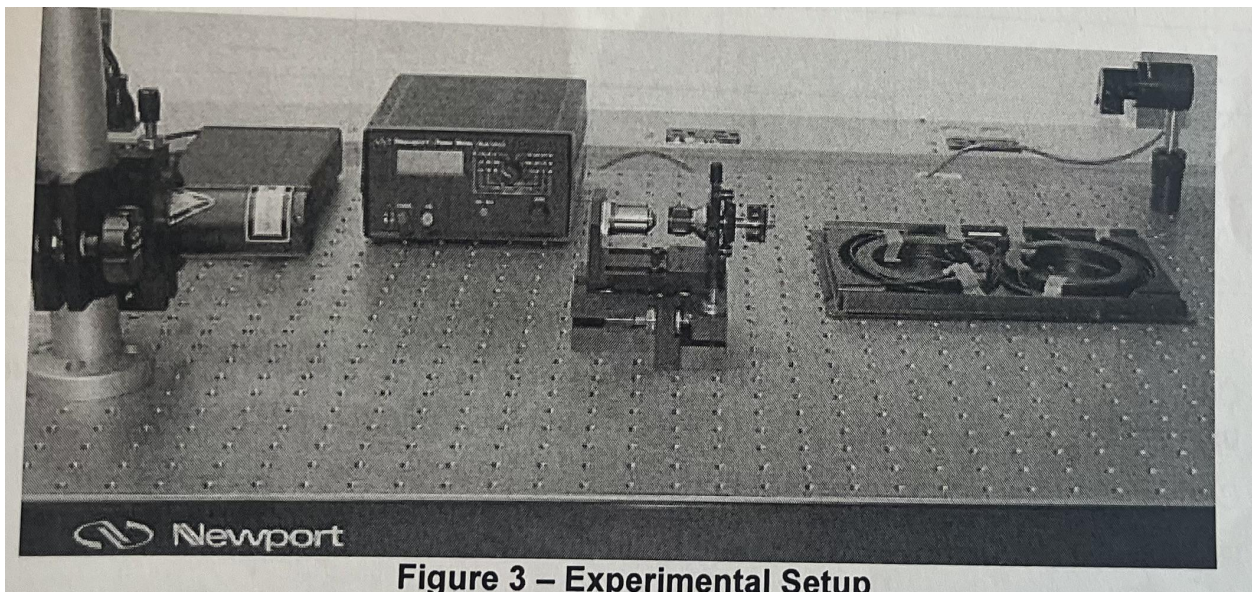


Figure 3 – Experimental Setup

1. Prepare one end of the fiber and place it in the chuck from the fiber coupler.
2. Mount one post holder on the optical table opposite of the laser power supply.
3. Attach the optical detector to a post and mount the assembly in the post holder.
4. Attach the bare fiber holder mount to the optical detector.

5. Prepare the other end of the fiber and place it in the bare fiber holder, attaching it to the optical detector.
6. Adjust the power meter using the “zero” knob until a 0 value appears on the display.
7. Unblock the laser.
8. Use adjustments from the fiber coupler to obtain the maximum output power on the power meter.
9. Record the output power.
10. Block the laser.

Multimode Bidirectional 2X2 Coupler:

1. Strip and cleave all fiber arms from ports 1,2,3, and 4 of the bidirectional coupler.
2. Use masking tape to mark and lightly tape down the fiber arms from ports 1,2,3, and 4 such that their end faces protrude slightly over the edge of the plastic casing.
3. Remove the fiber end from the bare fiber holder.
4. Unblock the laser.
5. Measure the optical power from each arm, ports 2,3, and 4. Record your results.
6. Block the laser.
7. Calculate the insertion losses, directivity loss, return loss, coupling ratio, and excess loss.
8. Convert power measurements to dBm.

$$\text{dBm} = -10 \log \left[\frac{P}{1 \text{ mW}} \right]$$

Where ***P*** is the power measured in mW.

Lab setup

Making sure the fiber positioned was properly aligned with the HeNe laser.

Stripping and cleaving the fiber and making sure the laser is aligned.

Measuring power out.

Setting up the mode scrambler.

Switching power meters because the gray one died.

Setting up mechanical splice and measuring power in and out.

5.) Data & Observations

Table 1

Port	Input Power Port 1	Output Power		CR (dB)	IL (dB)	DL (dB)	EL (dB)
		mW	dBm				
2	3.07mW	6.08nW	52.17	0.97 dB	-	57.03 dB	11.35 dB
3		0.18	7.45		12.32dB		
4		0.045	13.47		18.34 dB		

CR:

$$-10\log (P_3 / P_3 + P_4)$$

$$-10\log (0.18 * 10^{-3} / 0.18 * 10^{-3} + 0.045 * 10^{-3}) = 0.97 \text{ dB}$$

IL:

$$IL_3 = -10\log (P_3 / P_1)$$

$$-10\log (0.18 * 10^{-3} / 3.07 * 10^{-3}) = 12.32 \text{ dB}$$

$$IL_4 = -10\log (P_4 / P_1)$$

$$-10\log (0.045 * 10^{-3} / 3.07 * 10^{-3}) = 18.34 \text{ dB}$$

DL:

$$-10\log (P_2 / P_1)$$

$$-10\log (6.08 * 10^{-9} / 3.07 * 10^{-3}) = 57.03 \text{ dB}$$

EL:

$$-10\log (P_3 + P_4 / P_1)$$

$$-10\log (0.18 * 10^{-3} + 0.045 * 10^{-3} / 3.07 * 10^{-3}) = 11.35 \text{ dB}$$

$$\text{dBm} = -10\log(P / 1 \text{ mW})$$

$$-10\log(0.18 \text{ mW} / 1\text{mW}) = 7.45 \text{ dB}$$

$$-10\log(0.045 \text{ mW} / 1\text{mW}) = 13.47 \text{ dB}$$

$$-10\log(6.08 \text{ nW} / 1\text{mW}) = 52.17 \text{ dB}$$

The data obtained is as accurate as possible by measurement tools and the conductors of the experiment. Port 1 input power 3.07 mW, port 2 output power was 6.08 nW, port 3 output power was 0.18 mW, and port 4 output power was 0.045 mW. Errors occur if the laser is not aligned, the assembly was incorrect, or if the fiber is not stripped / cleaved correctly.

6.) Answers to Lab Questions

1) Record the power output, $P_{IN} = 3.07 \text{ mW}$.

7.) Conclusion

In conclusion, the losses of the fiber are affected by the stripping and cleaving of the fiber. The experiment yielded unexpected results because the group had difficulties with perfecting the cleave which affected our measurements. The cleave was imperfect so there was scattering of light at the end of the fiber chuck near the objective lens and detector head. We tried re-cleaving the fiber and double checking to make sure that it was cleaned with alcohol to remove excess jacket and glass fragments. Even though we did this, the losses were too high, meaning that more improvements need to be implemented with cleaving. Additional improvements would be to ensure that user errors and equipment errors are at a minimum so that the results are as accurate as possible for future work.